

Q3) A mobile OS has the following Tasks.

<u>TASK</u>	<u>AT</u>	<u>BT</u>
Music Player	0	20
Touch event	3	1
Whatsapp Notification	5	2
Camera launch	6	4

↳ Using this data answer the following.

- 1) which scheduling algorithm's most suitable.
- 2) Draw the GT for SRTF.
- 3) Why selected Algorithm from (1) improves user experience.
- 4) which task receives the fastest response.

Sol

1) SRTF - Shortest Remaining Time First

Because this is a mobile OS, short interactive tasks such as:

- touch events
 - Notifications
 - camera launch
- ⇒ should receive CPU quickly rather than waiting behind a long task like the music player.

2) Ready Queue (RQ)

0:-

Music Player

 → only Process available → selected

3:-

Music Player	Touch event
--------------	-------------

 → MP remaining = 17
TE = 1
 $1 < 17 \rightarrow$ Touch selected

4:-

Touch Event	Music Player
------------------------	--------------

 → only remaining Process

5:-

Music Player	Whatsapp Notification
--------------	-----------------------

 → MP remaining = 16
whatsapp = 2
 $2 < 16 \rightarrow$ whatsapp selected

6:-

Whatsapp Notification

 → Whatsapp remaining = 1
camera = 4, Music = 16
 $1 < 4 < 16 \rightarrow$ whatsapp continues

7:-

Whatsapp Notification	Camera Launch
----------------------------------	---------------

 → Camera = 4 < Music = 16

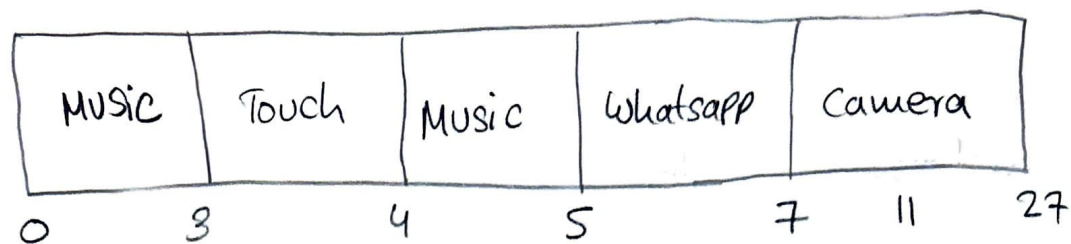
11:-

Camera Launch	Music Player
--------------------------	--------------

 → only Process remaining

27 :- DONE

Grantt chart (GT)



3)

↳ SRTF gives CPU quickly to short and interactive tasks, reducing their response time and making the mobile OS more responsive.

Short tasks → Quick Response → Better user experience

4) $RT = \text{First CPU Start Time} - AT$

<u>Task</u>	<u>AT</u>	<u>First Start</u>	<u>RT</u>
Music Player	0	0	0
Touch event	3	3	0
Whatsapp Notification	5	5	0
camera Launch	6	7	1

⇒ Music Player, Touch event & whatsapp notification receive the fastest response.